Week 6 Robot Car Design Review

AAI05 | 2 October 2026

The robot uses a Pico W and Robo Pico to follow the line, decode barcodes and execute encoder-controlled manoeuvres. It also estimates hump height, avoids obstacles and publishes telemetry. Software uses C, the Pico C SDK, micro T-Kernel and BARR-C:2018. Timing and calibration values are unmeasured test targets. Validate the electrical assumptions before full integration.

GitHub repository: https://github.com/phrypy/AAI05-robot-car

1 Team ownership

Member	Subsystem responsibility
You Yan	Mission, line following, barcode and integration
Daniyal	Motor drive, encoders, ultrasonic scan and servo avoidance
Patrick	IMU calibration, hump detection and height estimation
Brian	WiFi, MQTT telemetry, commands and recovery

Integration may be shared.

2 Functional requirements and acceptance checks

ID	Required behaviour	Acceptance test
FR1	Follow the 18 mm line using two analog IR sensors.	Complete the route; record line losses and recoveries.
FR2	Decode the adjacent barcode; reject unknown / incomplete reads.	Test the supplied symbols across the speed range and confirm their commands.
FR3	Execute left, right, straight and U-turn commands.	Measure distance, turn error and line reacquisition.
FR4	Detect obstacles, run coarse/fine scans and estimate location, width, closest point and left/right clearance. Detour and regain the line.	Compare with known obstacle positions and widths; stop before contact and regain the line.
FR5	Estimate hump height in mm and retain the highest valid height.	Test known-height humps. Record height error and invalid estimates. The estimator is unvalidated.
FR6	Publish state, wheel speed/counts, IR status, barcode, hump height, obstacle profile and travelled distance over MQTT. Accept validated start, stop and configuration commands; exclude remote steering.	Check broker logs and commands; test disconnect/reconnect.
FR7	Stop on explicit STOP, critical sensor fault, command expiry or recovery timeout.	Inject faults in each active mission state and measure stop response.

3 Nonfunctional requirements

ID	Target	Validation
NFR1	Motion update 10 ms; line ADC 5 ms; IMU 20 ms.	Record jitter and missed deadlines under full load.
NFR2	Apply a latched STOP within one 10 ms Motion cycle; detection adds latency. Motion expires targets after 100 ms.	Measure latency from STOP request to driver brake output.
NFR3	Control operates without WiFi. Network waits are bounded.	Disconnect WiFi and stall the broker during navigation.
NFR4	Forward ping interval while moving: 65–70 ms. During stopped scans: ≥65 ms between pings, after servo settling. ECHO timeout: 30 ms. Stop requested motion if forward-data age exceeds 150 ms.	Measure with a logic analyser; set settling time from the actual servo.
NFR5	Critical-event queue or capture-buffer overflow latches a fault and stops motion. Routine telemetry may be dropped with a counter. Stale critical data or expired targets stop motion.	Stress queues and capture buffers, remove critical data and check the fault latch and drop counters.

STOP response measures the time from the request to the driver brake output. Physical stopping distance is a separate measurement. Test timing with sensing and WiFi running together.

Hardware inventory and test equipment

Equipment	Quantity or scope	Role in the design
Pico W and Robo Pico	Mounted controller and carrier	Controller, C SDK, RTOS, radio and motor drivers
Geared motors with encoders	Two mounted wheel assemblies	Independent left/right drive and distance feedback
IR reflective sensors	Three modules shown in supplied photos	Two analog line sensors and one digital barcode sensor
GY-511 and ultrasonic sensor	One each	Acceleration/magnetic field and ultrasonic range
Micro servo	One mounted servo	Ultrasonic scan actuator; mount the sensor before testing
USB power bank and cable	5 V output confirmed; current rating pending	System power through Pico USB
Debug probe and cables	Development equipment	Programming and debugging through dedicated SWD
Chassis, wheels and caster	Two rubber wheels and one all-direction caster	Mechanical support and differential-drive movement
Mounting parts, cables and tools	Plates, screws, nuts, spacers, screwdriver and nozzle spanner	Assembly and wiring; no signal GPIOs
Combo tiles	Two sets	Track assembly and repeatable tests

The line is 18 mm wide, with a 20 mm gap to the barcode. Barcode length is approximately 141 mm; its nearest edge is 29 mm from the line centre. Place the third IR sensor within the stripe region. Check stripe width, side, mounting height and speed on the track. The A/Z sample does not define the full command alphabet, so keep the mapping configurable.

Confirm the letter-to-action mapping before command acceptance tests.

4 Hardware integration schematic

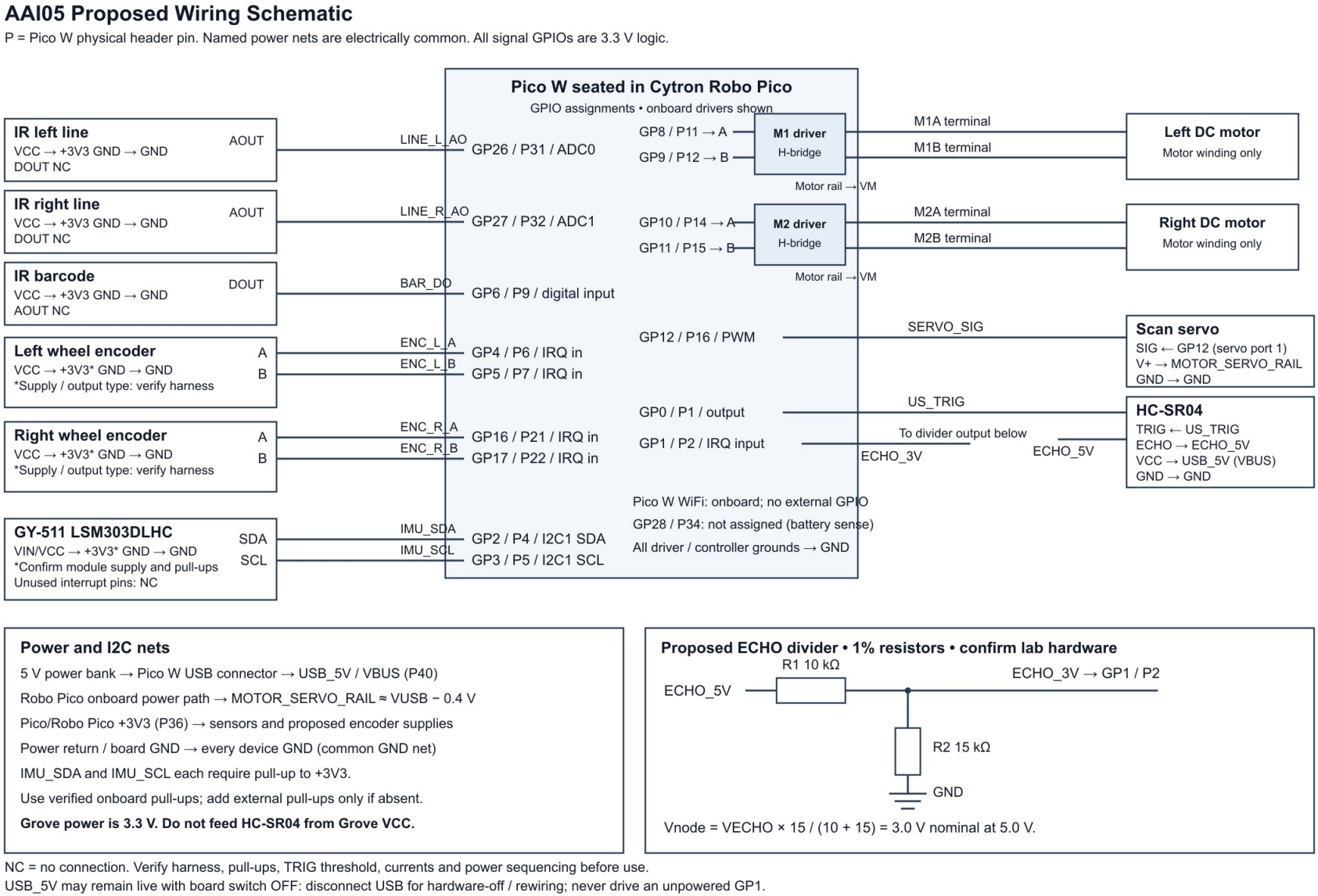


Figure 1 Component wiring. Matching net labels identify electrically common connections. P indicates a physical Pico header pin. The connection table lists assembly locations; the debug drawing shows development connections.

4.1 Connection locations and pin justification

Connect external signals through the labelled Robo Pico GPIO breakout. Grove ports with the same GPIO share one electrical connection. Do not connect a second device to those shared signals.

Component signal	GPIO and physical pin	Connection location	Pin choice and owner
Left motor driver A / B	GP8 P11 / GP9 P12	Internal driver; motor to M1A/M1B terminals	Fixed Robo Pico wiring; PWM slice 4. Motion control.
Right motor driver A / B	GP10 P14 / GP11 P15	Internal driver; motor to M2A/M2B terminals	Fixed wiring; PWM slice 5. Motion control.
Left encoder A / B	GP4 P6 / GP5 P7	GP4/GP5 breakout; same nets as Grove 3	Two digital edge inputs for quadrature; Encoder ISR / Motion.
Right encoder A / B	GP16 P21 / GP17 P22	GP16/GP17 breakout; same nets as Grove 4	Independent quadrature inputs; Encoder ISR / Motion.
Left IR AO	GP26 P31 / ADC0	GP26 breakout; same net on Grove 5 and 6	Analog reflectance for steering. Line and barcode.
Right IR AO	GP27 P32 / ADC1	GP27 breakout; same net as Grove 6	Second independent analog sample. Line and barcode.
Barcode IR DO	GP6 P9	GP6 breakout; same net as Grove 5	Digital comparator edge capture. Barcode ISR / Line and barcode.
GY-511 SDA / SCL	GP2 P4 / GP3 P5	GP2/GP3 breakout; also Grove 2 / Maker	I2C1 pair; sole bus owner is IMU and hump.
HC-SR04 TRIG / ECHO	GP0 P1 / GP1 P2	GP0/GP1 breakout; also Grove 1	TRIG output and conditioned ECHO edge input. Obstacle and servo.
Servo SIG	GP12 P16	Robo Pico servo port 1	PWM slice 6, separate from motor PWM. Obstacle and servo.
Sensor supply and return	3V3 P36 / GND P38	Labelled 3V3 and GND breakout	Common return and planned 3.3 V sensor supply. Confirm GY-511 terminal and encoder compatibility.
Ultrasonic supply	VBUS P40 / GND	VBUS breakout and common ground	USB 5 V net; use the reviewed lab interface shown in Figure 1.
WiFi radio	Onboard; no external GPIO	Pico W internal radio connection	WiFi and MQTT task; no external pin assignment.
Debug probe SC / SD / GND	SWCLK / SWDIO / GND	Dedicated Pico W SWD connector	Debug/programming interface; no extra signal GPIO.

Leave line-sensor DO, barcode-sensor AO and GY-511 interrupt pins disconnected. GP28 is unassigned because the battery-sense jumper may connect it to a voltage divider. GP18, GP20, GP21 and GP22 serve onboard functions outside this design. Keep the Pico W pins reserved for WiFi free.

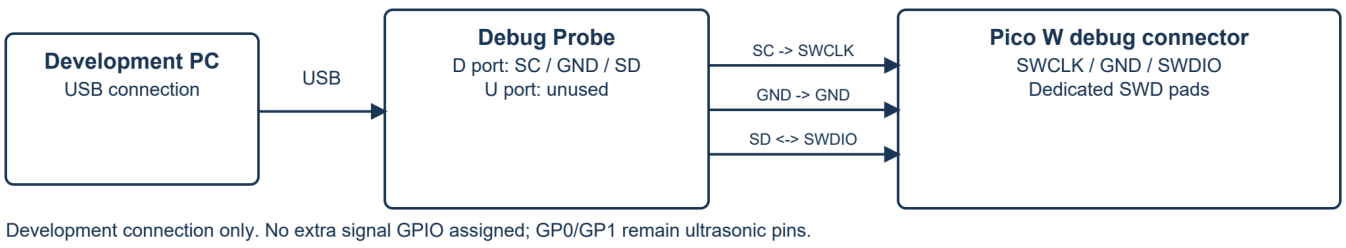


Figure 2 Debug probe connection. Probe UART is unused. GP0/GP1 are assigned to ultrasonic TRIG/ECHO.

Encoder A/B and supply labels identify functions; connector orientation still needs checking. Record the GY-511's printed supply label. Do not confuse its power input with a regulator output.

Motor inputs A/B select 00 brake, 10 forward, 01 reverse and 11 coast. Wheel direction depends on motor polarity. Only Motion writes motor outputs; only the Obstacle and Servo task writes servo PWM. Motors use PWM slices 4 and 5, while GP12 uses slice 6. Grove 2 and Maker share GP2/GP3; GP26 on Grove 5/6 is one signal. GP28 is unassigned.

Unplug the USB power bank before rewiring. Raw VBUS may remain live with the board switch off; check that ECHO cannot drive an unpowered GP1. Before applying power, check the encoder harness and output levels, GY-511 supply and pull-ups, TRIG threshold, ECHO divider and load current, including motor stall and servo startup. Board limits are 1 A continuous per motor channel, 1.5 A peak for <5 s, motor PWM ≤20 kHz and total Grove +3V3 current ≤300 mA. The divider and USB-derived rail values are design values, not measurements.

5 RTOS tasks and ownership

Task / handler	Trigger	Priority order*	Ownership and data path
Encoder / barcode / ECHO ISR	GPIO edge	Interrupt	Encoder ISR updates counts; barcode ISR records edge/time/count snapshots; ECHO ISR records pulse timestamps. Capture is bounded. Overflow latches a fault.
Motion control	10 ms + stop wakeup	1 highest	Sole GP8-GP11 writer. Reads counts and timestamped targets, checks STOP and 100 ms expiry, and publishes speed, distance and standstill status.
Mission supervisor	20 ms / event	2	Controls mission state; sends wheel targets, watchdog updates and scan requests.
Line and barcode	5 ms ADC / edges	3	Reads GP26/27 ADC, GP6 captures and encoder distance. Sends latest steering to Mission and symbol/fault events to the queue.
Obstacle and servo	65–70 ms forward / scan	4	Uses GP0 for TRIG, GP1 for ECHO capture and GP12 for servo control. Waits for servo settling, then runs coarse/fine scans to produce a profile.
IMU and hump	20 ms	5	Sole I2C1 owner. Uses calibrated acceleration and encoder distance to estimate height, peak and validity.
WiFi and MQTT	100 ms / event	6 lowest	Owns the network stack. Publishes copied status and validates commands.

*The order ranks task urgency; it does not specify micro T-Kernel priority numbers. Confirm the port's priority rules before assigning them. Motion has the highest task priority to meet output deadlines. Mission follows, then the short sensor tasks. Network work runs last and uses bounded waits. Check execution time and jitter under load. Interrupt priority is separate.

6 Task interactions and resource rules

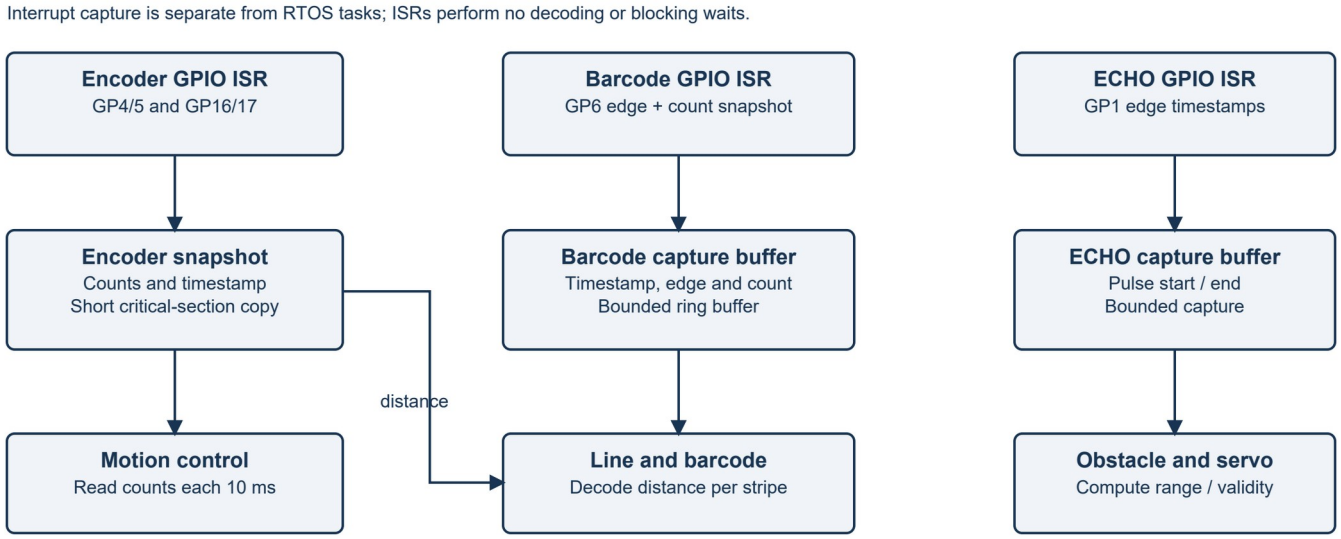


Figure 3 Interrupt capture paths. Each barcode edge records an encoder-count snapshot. The Line and Barcode task converts count differences to stripe distance. Figure 4 shows the distance snapshots used by the IMU estimator.

6.1 Control and communication paths

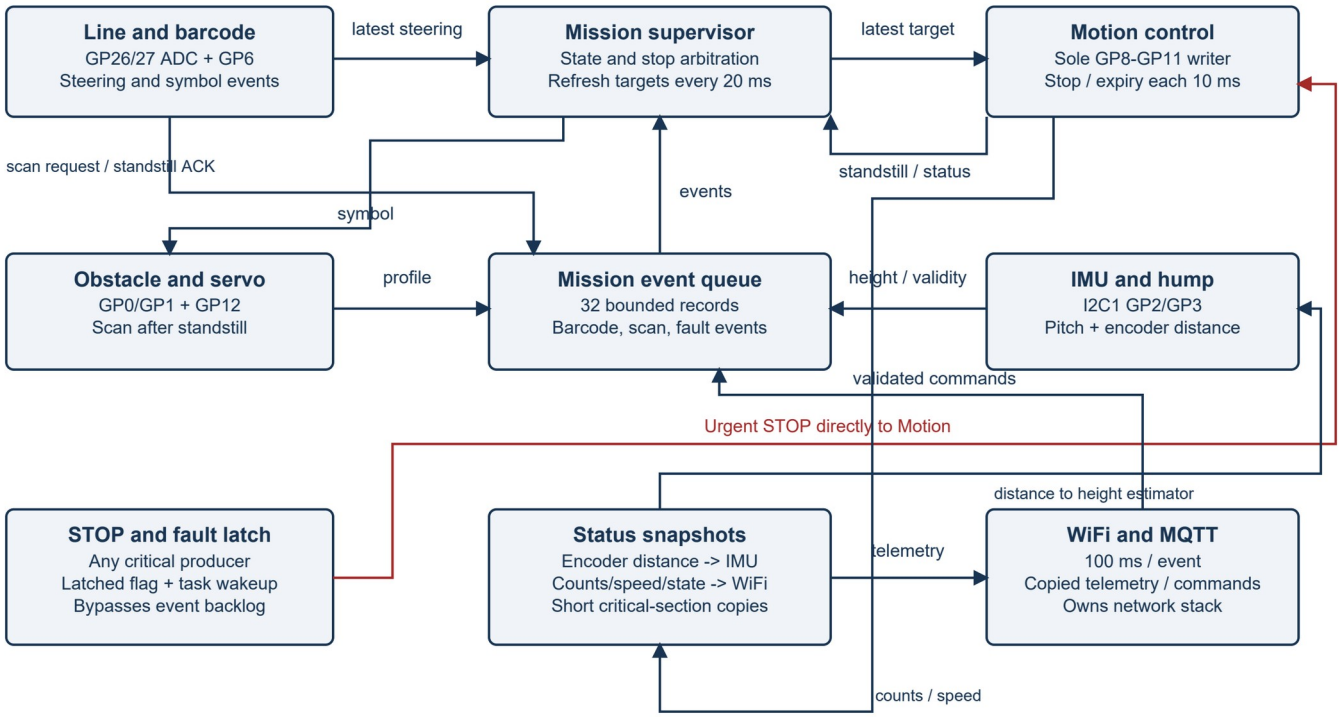


Figure 4 Task and object interactions. Steering and target slots hold the latest record; new values replace old ones. The event queue holds discrete events. Encoder snapshots feed barcode decoding and hump estimation. Telemetry uses copied status.

An encoder snapshot holds left/right counts and a timestamp. Copy it atomically in a short critical section. At each GP6 edge, the barcode ISR records the counts to tie stripe width to travel distance. IMU and Hump reads timestamped distance increments for height estimation. Motion publishes speed, distance and standstill status for Mission and telemetry.

Software object	Rule
Critical event queue	32 fixed-size records containing source, type, timestamp and payload. Producers do not block. A full queue latches STOP. Telemetry uses a separate copied snapshot.
Steering and target slots	Hold one latest-value record each, protected by short critical sections. Targets carry a timestamp and 100 ms expiry. Mission refreshes them every 20 ms during motion.
STOP flag and wakeup	A latched fault/STOP flag bypasses the queue. Wake Motion with a primitive safe for the calling ISR or task. Motion checks the flag each cycle.
Capture buffers	Barcode/ECHO records are bounded; overflow is a critical fault. ISRs only capture data and update counts/status.
Shared resources	IMU owns I2C1; WiFi owns the network stack; Motion owns motor PWM; Obstacle owns servo PWM. A bus with one task owner needs no mutex.

Use the micro T-Kernel objects supported by the selected port. Check ISR-safe APIs, wakeup behaviour and numeric priority rules before implementation. Never wait for I2C, ECHO, servo settling or networking inside a critical section.

An obstacle requests a temporary brake through Mission. Scan starts only after Motion acknowledges measured standstill within the timeout. Keep the servo forward while moving. Run coarse/fine scans while braked, then settle the servo forward and obtain a fresh valid range before resuming. A successful detour needs no fault reset. Explicit STOP and critical faults stay latched.

Calculate the stop threshold from maximum speed, worst-case sensing and output latency, measured braking distance and clearance margin. It must remain within reliable sonar range. The current policy treats no echo as invalid; test weak returns and open space. Check barcode distance resolution at the narrowest stripe and highest speed.

7 System and subsystem statecharts

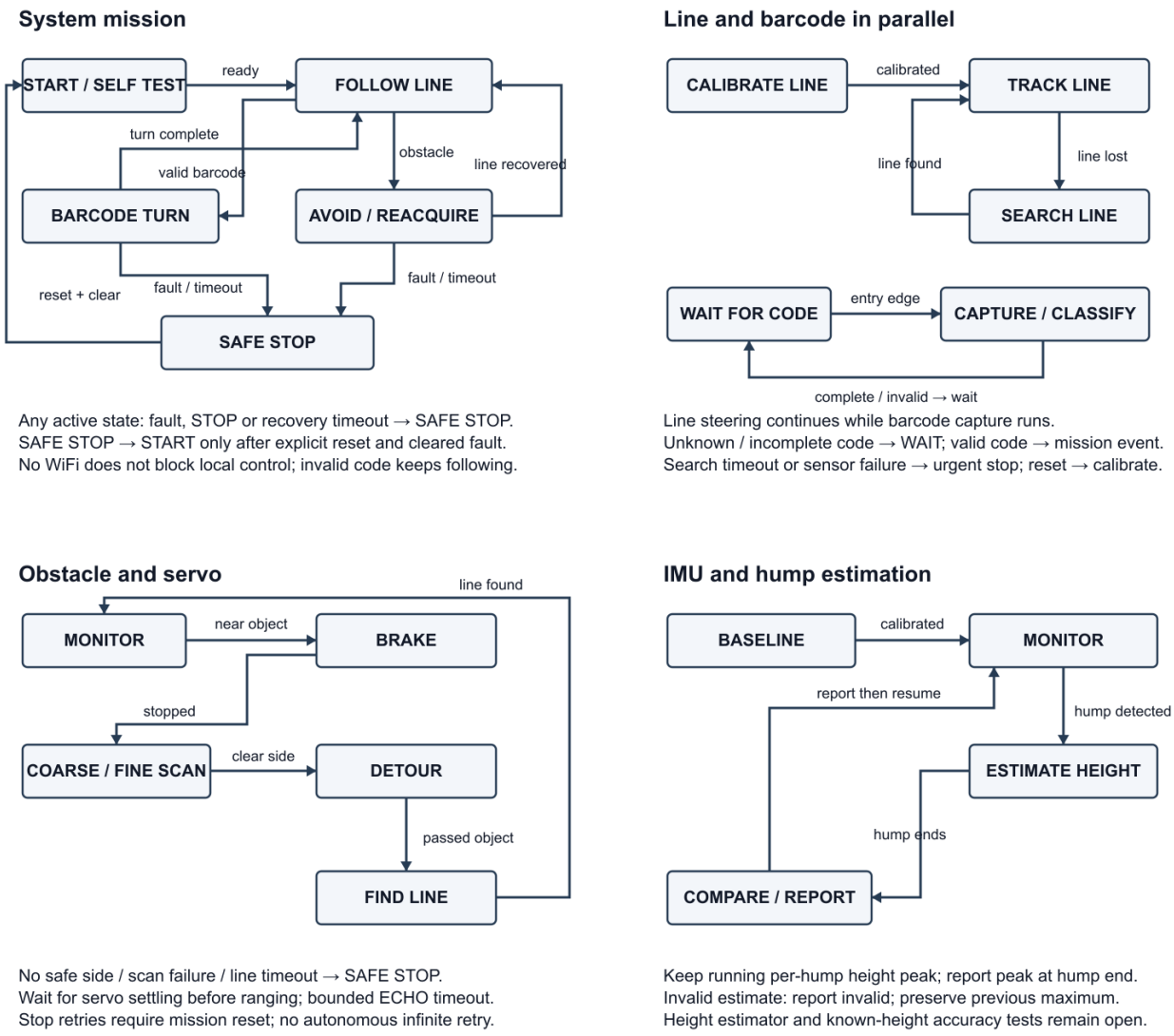


Figure 5 System and three subsystem statecharts. Line tracking continues during barcode capture. STOP and faults override movement.

Transitions use events or bounded conditions. Mission maintains steering during barcode capture and handles STOP/faults before turns or detours. Encoder manoeuvres finish when their targets and recovery conditions are met. Stale sensor data or a turn/search timeout latches SAFE STOP. Reset requires fault clearance and a new self-test.

7.1 Subsystem purpose and task mapping

Subsystem	Related tasks	Statechart purpose
Line and barcode	Line and barcode; Mission; barcode ISR	Keep tracking during barcode capture. States define calibration, search, capture completion, invalid-code rejection and recovery.
Obstacle and servo	Obstacle and servo; Mission; Motion; ECHO ISR	Sequence brake, standstill, coarse/fine scan, detour and line reacquisition. Scan begins only after movement stops.
IMU and hump	IMU and hump; encoder snapshot producer	Track baseline, hump entry/exit, height peak and reporting. Reject invalid estimates when updating the maximum.

Line and barcode: calibrate, then track. Line loss starts a search with a timeout. A valid barcode sends an event to Mission while tracking continues; unknown or incomplete codes are rejected. Search timeout or critical failure latches STOP.

Obstacle and servo: request brake when an obstacle is near. At standstill, scan from settled servo positions. A clear side permits a detour and line search, each with a timeout. No safe side, capture failure or recovery timeout latches STOP. A successful detour needs no fault reset.

The GY-511 measures acceleration and magnetic field. It has no gyroscope. Estimate height from gravity-referenced pitch and encoder path increments: $\Delta h \approx \Delta s \sin \theta$. Calibrate the mounting frame and ground baseline, and use a speed range where dynamic acceleration is controlled. Accumulate $h[k] = h[k-1] + \Delta s \sin \theta$. Keep the running peak for each hump and report it at hump end; the integral after descent may be near zero. Update the global maximum only from valid peaks. This method assumes chassis pitch follows the direction of travel. Wheelbase and contact geometry can break that assumption on short humps, while slip, dynamic acceleration and tilt error can bias the result. Compare the reported reference point with known-height humps to establish accuracy. Until validation passes, mark height estimates invalid.

Before an autonomous run, measure and set scan angles, servo limits and settling time, obstacle and freshness thresholds, line/turn timeouts, clearance margin and height-error tolerance. These settings and the hardware still need validation.

8 Assumptions and planned validation

ID / owner	Assumption and risk	Validation / status
A1 / Daniyal	M1 is left, M2 right; encoder supply and A/B outputs are 3.3 V compatible. Wrong harness/direction can damage inputs or reverse control.	Inspect harness/module; low-speed polarity/count test. Open.
A2 / You Yan	Two AO sensors can track while side DO sensor reads all required symbols. Wrong spacing/mapping causes losses or incorrect turns.	Printed track, black/white calibration, speed/stripe test and mapping confirmation. Open.
A3 / Patrick	GY-511 supply/pull-ups and the pitch/encoder estimator are suitable. Dynamic acceleration or slip may bias height.	Inspect module, log baseline/dynamics; compare known-height ramps; set tolerance. Open.
A4 / Brian	The power bank can supply the load, and WiFi/MQTT recovery does not block control. Brownout or blocked updates may stop control.	Calculate the load budget and measure rail voltage/current under combined load. Stress-test broker outages. Open.
A5 / Daniyal	HC-SR04 TRIG accepts 3.3 V and ECHO divider yields safe timing levels. Wrong levels or stale scan can damage GPIO or misroute car.	Inspect lab circuit; scope TRIG/ECHO, >60 ms spacing and switch-off back-power. Disconnect USB for hardware-off; prevent ECHO driving an unpowered GP1. Open.
A6 / Team	The pin map and debug connector locations match the installed boards. Shared nets may cause conflicts.	Check board labels, GP28 sense jumper and SWD connector; compare final harness map with Figure 1 and Section 4.1. Open.

9 AI design evidence

9.1 Member AI discussions

Member	Discussion and status	Useful idea and limitation	Decision or follow-up
You Yan	Asked AI to explain the robot project and suggest a pin map. Also asked it to compare the report with the Week 6 template and address the issues found.	Useful: fixed motor pins, one writer per actuator and explicit data paths. Limitation: a pin map cannot verify harness order, barcode meanings or height accuracy.	Keep the fixed motor pins and sensor data paths, with one task owner per actuator. Check harness order, barcode mapping and height accuracy during integration.
Daniyal	Asked Gemini how the components work together to understand the project.	Useful: clarified how the components interact. Limitation: practical understanding still requires assembling and observing the hardware.	Used this understanding to contribute to schematic and wiring choices.
Patrick	tbc	tbc	tbc
Brian	Asked AI to review WiFi and MQTT telemetry, command handling and connection recovery for the robot.	Useful: prevent network delays from blocking local control and validate incoming commands. Limitation: connection recovery and timing still require hardware testing.	Use copied status for telemetry and pass validated commands to Mission. Test WiFi disconnection and broker outages while the car runs.

9.2 Team architecture decisions

Idea	Team decision	Engineering reason	Affected artefacts
Use Robo Pico onboard drivers; M1 left / M2 right	Accepted	Use the onboard drivers and fixed GPIOs. Keep M1 left and M2 right throughout the design. Check motor polarity and encoder direction during initial tests.	FR3, schematic, task table, A1
Two analog line sensors and side digital barcode sensor	Accepted with validation	Two analog sensors correct line position; the separate digital sensor captures barcodes while tracking continues. Test mounting, black/white calibration and capture across the speed range.	FR1/2, pin map, A2
Single motor and servo task owners	Accepted	Motion Control alone writes the motor outputs; the Obstacle and Servo task alone writes the servo output. Other tasks send requests. This avoids conflicting commands and gives each actuator one output owner for stop testing.	Sections 5/6 and statecharts
Telemetry queue overload always stops movement	Modified	Drop routine telemetry and count losses when communication is congested. Critical-event queue or capture-buffer overflow stops motion: control information may have been lost.	NFR5 and resource rules
Add debug SWD and missing distance/capture paths	Accepted	Include SWD debugging and show how encoder snapshots feed barcode decoding, height estimation and telemetry. These paths connect the wiring diagram to the software dependencies.	Figures 2-4 and Section 4.1
Use pitch and encoder distance to estimate hump height	Accepted with validation	Use this initial method with the available sensors. Test known-height humps, accounting for chassis geometry, slip and acceleration. Set the error tolerance before testing. Revise and document the estimator if it fails.	FR5, height statechart, A3

9.3 AI review findings

Review request: Compare the report with the Week 6 template. Check the schematic, pin mapping, nonfunctional requirements, component coverage and agreement between the diagrams and tables.

Gap found	Response	Action and evidence
Critical and routine queue handling inconsistent	Fixed	NFR5 and Section 6 now distinguish critical events/capture from droppable telemetry.
Debug probe omitted from wiring	Fixed	Dedicated SWD drawing and pin-map entry added; probe UART unused.
Assembly connector locations unclear	Partially fixed	Breakout, servo and terminal locations are listed. Encoder harness order and GY-511 supply label still need the A1/A3 checks.
Encoder distance and capture paths absent	Fixed	Figures 3/4 and task table show barcode count capture, IMU distance and telemetry snapshots.
Subsystem metadata omitted	Fixed	Section 7.1 lists tasks, purpose and state/fault behaviour.
Barcode mapping and height accuracy unconfirmed	Needs validation	The barcode-to-action mapping is unconfirmed. A2 and A3 specify checks for barcode operation and height accuracy.

10 Requirement traceability

FR ID	Task and hardware path	Statechart or requirement link
FR1 / FR2	Line and barcode -> GP26/27/6 + encoder-distance snapshots	Track/search and barcode capture; NFR1/5
FR3	Mission -> latest target -> Motion -> GP8-GP11; encoder feedback	Barcode turn and line reacquisition; NFR2
FR4	Obstacle -> GP0/1/12; profile to Mission; Motion standstill ACK	Brake, scan, detour and find line; NFR4
FR5	IMU and hump -> I2C1 GP2/3 + encoder distance	Baseline, height estimate and peak report; NFR1
FR6	WiFi and MQTT -> copied status / validated events; onboard radio	Local control survives network loss; NFR3
FR7	Critical producers -> STOP latch + wakeup -> Motion brake	SAFE STOP; reset after cleared fault; NFR2/5